Lesson 11.1 - Hypothesis Testing - Difference of Two Means

> **Video:** Hypothesis Testing - Difference of Two Means - Student's t-Distribution & Normal Distribution by The Organic Chemistry Tutor **Chapter:** 11 - Hypothesis Testing - 2 Samples

Setup

Two **independent** groups. Does their population mean differ?

$$H_0 : \mu_1 = \mu_2 \;\; (\mu_1 - \mu_2 = 0) \qquad H_a : \mu_1 \neq \mu_2 \;\; (\mu_1 - \mu_2 \neq 0)$$

Test statistic

$$t \text{ (or } z) = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Under H_0, $\mu_1 - \mu_2 = 0$. The denominator is the **standard error of the difference**.

Sample sizes	Use
Small (< 30)	**t-distribution** (Welch), with the degrees of freedom below
Large (≥ 30)	**z** (normal distribution)

Degrees of freedom (Welch-Satterthwaite):

$$\text{df} = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(s_1^2/n_1\right)^2}{n_1 - 1} + \frac{\left(s_2^2/n_2\right)^2}{n_2 - 1}}$$

Example 1 - two teachers (t-test)

Do two teachers' classes score differently? $\alpha = 0.05$.

	Class 1	Class 2
n	15	12
$\bar{x}$	82	84
s	2.4	1.7

Hypotheses: $H_0 : \mu_1 - \mu_2 = 0$, $H_a : \mu_1 - \mu_2 \neq 0$ (two-tailed, $\alpha/2 = 0.025$).

Both samples are small → t-test.

Degrees of freedom:

- $s_1^2/n_1 = 5.76/15 = 0.384$ and $s_2^2/n_2 = 2.89/12 = 0.24083$
- Numerator: $(0.384 + 0.24083)^2 = (0.62483)^2 = 0.390417$

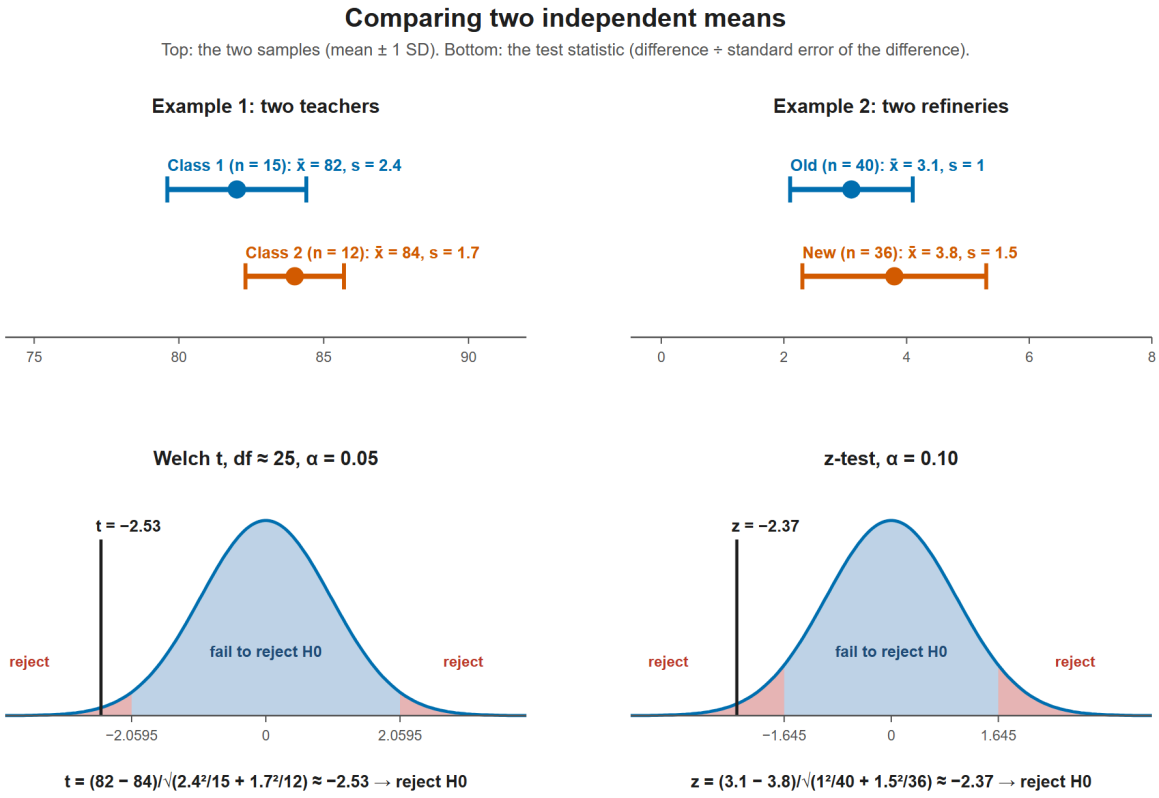
- Denominator: $\frac{0.384^2}{14} + \frac{0.24083^2}{11} = 0.010533 + 0.005273 = 0.015805$
- $\text{df} = 0.390417 / 0.015805 \approx 24.7$

The video rounds this to **25**, giving $t_{0.025,\,25} = 2.0595$. Many textbooks round df **down** to be conservative ($24 \to t = 2.064$). Both give the same decision here.

Test statistic:

$$t = \frac{82 - 84 - 0}{\sqrt{0.384 + 0.24083}} = \frac{-2}{0.79046} \approx -2.53$$

$-2.53 < -2.0595$, which is in the rejection region. **Reject H_0.** At the 5% level there is a significant difference between the two teachers' class means.



Top: the two samples in each example. Bottom: the difference in means divided by its standard error. In both examples the statistic lands in the left rejection region, so the means differ significantly.

Example 2 - two oil refineries (z-test)

Is there a difference in refining rate (liters per second)? $\alpha = 0.10$.

	Old factory	New factory
n	40	36
$\bar{x}$	3.1	3.8
s	1.0	1.5

Hypotheses: $H_0 : \mu_1 = \mu_2,\ H_a : \mu_1 \neq \mu_2$ (two-tailed).

Both samples are large → z-test. The confidence level is 90%, so the left area for the critical value is 0.95: $z = \pm 1.645$.

$$z = \frac{3.1 - 3.8}{\sqrt{\dfrac{1^2}{40} + \dfrac{1.5^2}{36}}} = \frac{-0.7}{\sqrt{0.025 + 0.0625}} = \frac{-0.7}{0.2958} \approx -2.37$$

$-2.37 < -1.645$ → **reject** H_0. At the 10% level there is a significant difference between the factories; the new one refines faster.

Key takeaways

- H_0: equal means (difference 0). Test statistic = (difference in sample means) ÷ $\sqrt{s_1^2/n_1 + s_2^2/n_2}$.
- Small samples → t with Welch's df. Large samples → z.
- Compare with the critical value(s), or with the p-value, as in the one-sample case.